

Titration report

LabX

Method

Method ID	CDC_GranTED	Release state	Released
Name	Titration with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/3/2026 3:42:13 PM		

Summary

Samples

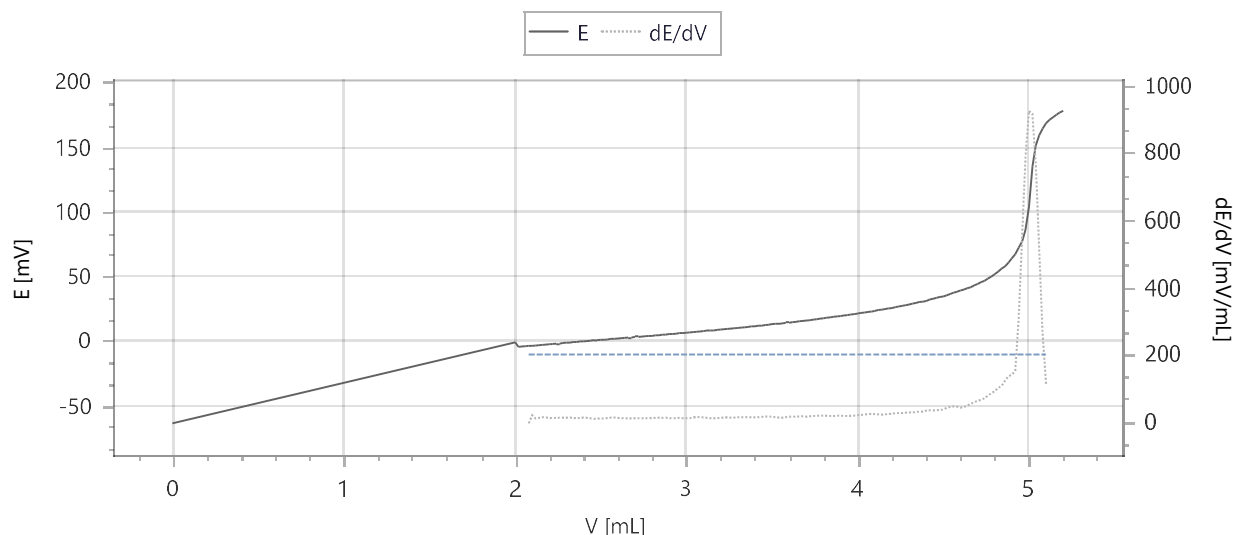
Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.007658 mL
		Concentration R2	0.10015 mol/L

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.007658 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10015 mol/L	OK

Titration (EQP) [1] Scope 1/1, Sample 1/1

E - V & dE/dV - V Curve Scope 1/1, Sample 1/1



Measured Values Scope 1/1, Sample 1/1

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-63.7		25.0			
6	2.0000	-1.1	2.0000	25.0	62.6		
40	2.0200	-4.6	0.0200	25.0	-3.5		

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42	2.0400	-4.3	0.0200	25.0	0.3	
46	2.0600	-4.1	0.0200	25.0	0.2	
49	2.0800	-3.8	0.0200	25.0	0.3	-4.50
52	2.1000	-3.9	0.0200	25.0	-0.1	19.71
55	2.1200	-3.6	0.0200	25.0	0.3	9.73
58	2.1400	-3.4	0.0200	25.0	0.2	11.74
61	2.1600	-3.1	0.0200	25.0	0.3	13.90
64	2.1800	-2.8	0.0200	25.0	0.3	13.29
67	2.2000	-2.6	0.0200	25.0	0.2	11.18
70	2.2200	-2.3	0.0200	25.0	0.3	11.61
73	2.2400	-2.1	0.0200	25.0	0.2	11.35
76	2.2600	-2.1	0.0200	25.0	0.0	12.28
79	2.2800	-1.6	0.0200	25.0	0.5	12.08
82	2.3000	-1.3	0.0200	25.0	0.3	12.16
86	2.3200	-1.2	0.0200	25.0	0.1	12.86
89	2.3400	-0.9	0.0200	25.0	0.3	11.79
92	2.3600	-0.8	0.0200	25.0	0.1	9.80
95	2.3800	-0.4	0.0200	25.0	0.4	11.88
98	2.4000	-0.2	0.0200	25.0	0.2	12.52
101	2.4200	-0.1	0.0200	25.0	0.1	12.00
104	2.4400	0.2	0.0200	25.0	0.3	10.86
107	2.4600	0.6	0.0200	25.0	0.4	8.81
111	2.4800	0.6	0.0200	25.0	0.0	9.73
114	2.5000	0.8	0.0200	25.0	0.2	10.05
117	2.5200	0.9	0.0200	25.0	0.1	9.87
120	2.5400	1.2	0.0200	25.0	0.3	11.28
123	2.5600	1.6	0.0200	25.0	0.4	12.58
126	2.5800	1.7	0.0200	25.0	0.1	12.80
129	2.6000	1.9	0.0200	25.0	0.2	11.53
132	2.6200	2.2	0.0200	25.0	0.3	11.02
135	2.6400	2.4	0.0200	25.0	0.2	10.16
138	2.6600	2.7	0.0200	25.0	0.3	10.48
141	2.6800	2.7	0.0200	25.0	0.0	10.02
144	2.7000	3.1	0.0200	25.0	0.4	10.51
147	2.7200	3.1	0.0200	25.0	0.0	10.69
150	2.7400	3.5	0.0200	25.0	0.4	10.80
153	2.7600	3.7	0.0200	25.0	0.2	10.48
156	2.7800	3.9	0.0200	25.0	0.2	10.44
160	2.8000	4.0	0.0200	25.0	0.1	11.31
162	2.8200	4.3	0.0200	25.0	0.3	11.48
166	2.8400	4.5	0.0200	25.0	0.2	12.29
169	2.8600	4.9	0.0200	25.0	0.4	11.99
172	2.8800	5.0	0.0200	25.0	0.1	12.08
175	2.9000	5.3	0.0200	25.0	0.3	10.96
178	2.9200	5.4	0.0200	25.0	0.1	10.96
181	2.9400	5.7	0.0200	25.0	0.3	11.02
184	2.9600	6.0	0.0200	25.0	0.3	11.77
187	2.9800	6.1	0.0200	25.0	0.1	10.64
190	3.0000	6.4	0.0200	25.0	0.3	10.78
193	3.0200	6.6	0.0200	25.0	0.2	10.92
196	3.0400	6.8	0.0200	25.0	0.2	13.02
199	3.0600	7.0	0.0200	25.0	0.2	14.67

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202	3.0800	7.4	0.0200	25.0	0.4	13.48
205	3.1000	7.7	0.0200	25.0	0.3	13.05
208	3.1200	8.0	0.0200	25.0	0.3	12.04
212	3.1400	8.1	0.0200	25.0	0.1	10.63
214	3.1600	8.2	0.0200	25.0	0.1	10.20
218	3.1800	8.6	0.0200	25.0	0.4	10.99
221	3.2000	8.8	0.0200	25.0	0.2	11.90
224	3.2200	9.0	0.0200	25.0	0.2	13.11
227	3.2400	9.3	0.0200	25.0	0.3	13.08
230	3.2600	9.5	0.0200	25.0	0.2	12.12
233	3.2800	9.8	0.0200	25.0	0.3	12.78
236	3.3000	10.1	0.0200	25.0	0.3	13.30
239	3.3200	10.3	0.0200	25.0	0.2	13.44
242	3.3400	10.5	0.0200	25.0	0.2	14.07
245	3.3600	10.9	0.0200	25.0	0.4	13.78
248	3.3800	11.2	0.0200	25.0	0.3	13.41
251	3.4000	11.4	0.0200	25.0	0.2	13.22
254	3.4200	11.7	0.0200	25.0	0.3	13.80
258	3.4400	11.9	0.0200	25.0	0.2	14.72
260	3.4600	12.2	0.0200	25.0	0.3	16.08
264	3.4800	12.6	0.0200	25.0	0.4	15.27
267	3.5000	13.0	0.0200	25.0	0.4	15.36
270	3.5200	13.2	0.0200	25.0	0.2	13.68
273	3.5400	13.4	0.0200	25.0	0.2	12.68
276	3.5600	13.6	0.0200	25.0	0.2	12.22
279	3.5800	14.1	0.0200	25.0	0.5	13.40
282	3.6000	14.1	0.0200	25.0	0.0	15.01
285	3.6200	14.6	0.0200	25.0	0.5	15.61
288	3.6400	14.8	0.0200	25.0	0.2	15.45
291	3.6600	15.2	0.0200	25.0	0.4	16.03
295	3.6800	15.5	0.0200	25.0	0.3	16.28
298	3.7000	15.8	0.0200	25.0	0.3	15.12
301	3.7200	16.1	0.0200	25.0	0.3	16.59
304	3.7400	16.5	0.0200	25.0	0.4	17.22
307	3.7600	16.7	0.0200	25.0	0.2	17.70
310	3.7800	17.2	0.0200	25.0	0.5	18.65
313	3.8000	17.6	0.0200	25.0	0.4	18.51
316	3.8200	17.9	0.0200	25.0	0.3	17.93
319	3.8400	18.2	0.0200	25.0	0.3	17.58
323	3.8600	18.7	0.0200	25.0	0.5	17.20
326	3.8800	18.9	0.0200	25.0	0.2	18.22
329	3.9000	19.3	0.0200	25.0	0.4	18.30
332	3.9200	19.7	0.0200	25.0	0.4	17.46
335	3.9400	20.1	0.0200	25.0	0.4	17.72
338	3.9600	20.4	0.0200	25.0	0.3	17.94
341	3.9800	20.7	0.0200	25.0	0.3	18.48
344	4.0000	21.1	0.0200	25.0	0.4	19.51
347	4.0200	21.6	0.0200	25.0	0.5	20.39
350	4.0400	21.9	0.0200	25.0	0.3	22.37
354	4.0600	22.5	0.0200	25.0	0.6	22.83
357	4.0800	22.8	0.0200	25.0	0.3	22.96
360	4.1000	23.3	0.0200	25.0	0.5	23.11

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363	4.1200	23.9	0.0200	25.0	0.6	22.47	
366	4.1400	24.2	0.0200	25.0	0.3	21.34	
369	4.1600	24.7	0.0200	25.0	0.5	22.31	
372	4.1800	25.1	0.0200	25.0	0.4	23.20	
376	4.2000	25.5	0.0200	25.0	0.4	24.73	
379	4.2200	26.1	0.0200	25.0	0.6	25.26	
382	4.2400	26.7	0.0200	25.0	0.6	26.06	
385	4.2600	27.2	0.0200	25.0	0.5	26.18	
388	4.2800	27.6	0.0200	25.0	0.4	27.27	
391	4.3000	28.1	0.0200	25.0	0.5	28.10	
394	4.3200	28.9	0.0200	25.0	0.8	27.80	
397	4.3400	29.3	0.0200	25.0	0.4	29.55	
400	4.3600	30.1	0.0200	25.0	0.8	30.08	
403	4.3800	30.5	0.0200	25.0	0.4	30.29	
406	4.4000	31.0	0.0200	25.0	0.5	33.20	
409	4.4200	32.0	0.0200	25.0	1.0	34.28	
412	4.4400	32.5	0.0200	25.0	0.5	34.06	
416	4.4600	33.3	0.0200	25.0	0.8	34.07	
419	4.4800	34.1	0.0200	25.0	0.8	35.45	
422	4.5000	34.4	0.0200	25.0	0.3	36.76	
425	4.5200	35.4	0.0200	25.0	1.0	41.70	
428	4.5400	36.2	0.0200	25.0	0.8	43.84	
432	4.5600	37.4	0.0200	25.0	1.2	45.81	
435	4.5800	38.0	0.0200	25.0	0.6	44.78	
438	4.6000	39.1	0.0200	25.0	1.1	42.35	
441	4.6200	39.8	0.0200	25.0	0.7	43.44	
444	4.6400	40.7	0.0200	25.0	0.9	48.66	
447	4.6600	41.6	0.0200	25.0	0.9	53.43	
450	4.6800	42.9	0.0200	25.0	1.3	58.22	
454	4.7000	44.1	0.0200	25.0	1.2	62.53	
459	4.7200	45.6	0.0200	25.0	1.5	65.53	
462	4.7400	46.5	0.0200	25.0	0.9	69.56	
465	4.7600	48.3	0.0200	25.0	1.8	78.10	
468	4.7800	49.6	0.0200	25.0	1.3	84.38	
472	4.8000	51.6	0.0200	25.0	2.0	90.90	
475	4.8200	53.6	0.0200	25.0	2.0	100.75	
481	4.8400	55.9	0.0200	25.0	2.3	106.35	
484	4.8600	57.5	0.0200	25.0	1.6	122.18	
487	4.8800	60.5	0.0200	25.0	3.0	135.27	
491	4.9000	63.7	0.0200	25.0	3.2	140.21	
494	4.9200	67.0	0.0200	25.0	3.3	150.98	
498	4.9400	72.1	0.0200	25.0	5.1	305.29	
505	4.9600	77.2	0.0200	25.0	5.1	550.00	
513	4.9800	85.8	0.0200	25.0	8.6	786.87	
527	5.0000	103.7	0.0200	25.0	17.9	924.15	
	5.007658	115.7				928.12	EQP
536	5.0200	135.1	0.0200	25.0	31.4	915.83	
541	5.0400	150.9	0.0200	25.0	15.8	765.10	
544	5.0600	159.0	0.0200	25.0	8.1	512.56	
548	5.0800	164.4	0.0200	25.0	5.4	259.02	
551	5.1000	168.3	0.0200	25.0	3.9	113.10	
555	5.1200	170.9	0.0200	25.0	2.6		

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558	5.1400	173.0	0.0200	25.0	2.1
561	5.1600	174.9	0.0200	25.0	1.9
564	5.1800	176.6	0.0200	25.0	1.7
567	5.2000	178.0	0.0200	25.0	1.4

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
03.02.2026 15:42:36	PredefinedUser	Administrator